

## Stacks (Assignment Solutions)

### Question 1 :

```
bool isPalin(string str) {  
    int n = str.size();  
    for(int i=0; i<n/2; i++) {  
        if(str[i] != str[n-i-1]) {  
            return false;  
        }  
    }  
  
    return true;  
}  
  
bool isPalindrome(ListNode* head) {  
    string str = "";  
    ListNode* temp = head;  
  
    while(temp != NULL) {  
        str += temp->val;  
        temp = temp->next;  
    }  
  
    return isPalin(str);  
}
```

### Question 2 :

```
string decodeString(string s) {  
    stack<int> st;  
    stack<string> st1;  
    string sb;  
    int n = 0;  
  
    for (char c : s) {  
        if (isdigit(c)) {
```

```
        n = n * 10 + (c - '0');
    } else if (c == '[') {
        st.push(n);
        n = 0;
        st1.push(sb);
        sb = "";
    } else if (c == ']') {
        int k = st.top();
        st.pop();
        string temp = sb;
        sb = st1.top();
        st1.pop();
        while (k-- > 0) {
            sb += temp;
        }
    } else {
        sb += c;
    }
}

return sb;
}
```

### Question 3 :

```
string simplifyPath(string path) {
    string ok;
    vector<string>store;
    int i = 1, n = s.size();
    while(i < n)
    {
        while(s[i]=='/') i++;
        while(s[i] and s[i] != '/') ok += s[i++];

        if(ok == ".." and store.size()) store.pop_back();
        else if(ok.size() and ok != "." and ok != "..")
store.push_back(ok);
        ok.clear();
    }
}
```

```
    ok = "/";  
    for(auto str:store) ok += str, ok += "/";  
    if(ok.size() > 1) ok.pop_back();  
  
    return ok;  
}
```

**Question 4 :**

```
int trap(vector<int>& height) {  
    int left = 0;  
    int right = height.size() - 1;  
    int leftMax = height[left];  
    int rightMax = height[right];  
    int water = 0;  
  
    while (left < right) {  
        if (leftMax < rightMax) {  
            left++;  
            leftMax = max(leftMax, height[left]);  
            water += leftMax - height[left];  
        } else {  
            right--;  
            rightMax = max(rightMax, height[right]);  
            water += rightMax - height[right];  
        }  
    }  
  
    return water;  
}
```